TARNDEEP SINGH

Computer Science & Data Science Undergraduate | B.Tech 2027

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PROFESSIONAL SUMMARY

B.Tech Computer Science and Engineering student specializing in Data Science at Akal University (CGPA: 9.02/10), with hands-on experience developing machine learning pipelines, computer vision systems, and full-stack applications. Experienced in Python, Java, data analysis, model evaluation, and integrating AI capabilities into practical applications.

TECHNICAL SKILLS

Programming: Python, Java, SQL

Machine Learning: Machine Learning, Deep Learning, Computer Vision, Classification, Feature Engineering, Data Preprocessing, Data Augmentation, Model Evaluation

Libraries: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, MediaPipe, Matplotlib, Seaborn, Plotly

PROJECTS

Indian Sign Language (ISL) Gesture-to-Text Recognition GitHub Ongoing

- Developing a real-time recognition prototype for 50 ISL signs and one idle class using the AI4Bharat INCLUDE-50 dataset (943 videos used).
- Built a landmark-based sequence pipeline using MediaPipe Pose and Hand Landmarkers, and implemented a Conv1D-BiLSTM-Multi-Head Self-Attention model with approximately 0.88 million parameters.
- Developed inference components using FastAPI, WebSockets, and ONNX Runtime; 12 automated tests pass.

Technologies: Python, PyTorch, MediaPipe, OpenCV, FastAPI, WebSockets, ONNX Runtime

StudySphere – Personalised AI Study Companion Live Demo | GitHub Dec. 2025

- Developed a student productivity platform integrating subject and topic tracking, notes, deadlines, attendance, focus sessions, and habits.
- Implemented a retrieval-based AI tutor that ranks relevant passages from student notes and generates evidence-grounded answers with citations.
- Designed local-first storage with optional Supabase authentication, PostgreSQL Row Level Security, and serverless AI integration.

Technologies: JavaScript, HTML, CSS, Node.js, Supabase, PostgreSQL, Vercel Functions, PDF.js

SONAR Rock vs. Mine Classification Live Demo | GitHub Sep. 2024

- Developed a binary classification pipeline using 208 sonar samples and 60 frequency-band features, including data cleaning, exploratory analysis, and spectral feature engineering.
- Implemented comparison of 11+ model configurations using nested cross-validation, hyperparameter tuning, and hold-out evaluation with ROC-AUC, F1, MCC, accuracy, and mine recall.
- Built a Streamlit prediction interface with threshold-based decisions and low-confidence review flags.

Technologies: Python, Scikit-learn, Pandas, NumPy, Streamlit

EDUCATION

Akal University, Talwandi Sabo, Punjab Aug. 2023 – Jun. 2027

B.Tech – Computer Science and Engineering (Data Science) **CGPA: 9.02/10**

RELEVANT COURSEWORK

Software Engineering, Operating Systems, Computer Networks, Data Handling and Visualization, Probability and Statistics, Artificial Intelligence, Natural Language Processing, Cyber Security

LEADERSHIP AND VOLUNTEERING

Founding President – DataNinjas, Akal Tech Society Nov. 2024 – Dec. 2025

- Helped establish and grow a 75-member technical community, promoting peer learning and student engagement.
- Managed and guided a team of 10 event volunteers while organising a quiz (48 participants), debate (25 participants), and treasure hunt (90 participants across 30 teams).

Student Learning Team Lead Feb. – May 2025

Led a group of 30 students, monitored participation in daily 30-minute English-learning sessions, and encouraged consistent engagement through follow-ups and individual motivation.

Vice President – FutureLens, Akal Tech Society Apr. 2026 – May 2026

Coordinated debate and presentation competitions in collaboration with club leadership and communication teams.

Conference Volunteer – CI2A 2026, Akal University Apr. 2026

Supported session management, documentation, and coordination with faculty and researchers.

ACHIEVEMENTS

- Selected for the internal evaluation round of Smart India Hackathon (SIH) 2025.
- Secured **1st position** in a cybersecurity quiz among 10 participating teams.
- Qualified through two rounds of the CodeForge Club individual coding hackathon (2026).
- Led TechTrail '26, coordinating a team of 5 members and organising a campus-wide treasure hunt involving 99 participants across 33 teams.